



Name: _____

Student Activity Sheet (SAS)

Date: _____

Course Name: **Information Assurance and Security 1**

Module Number: 09

Topic	Secure Data Storage
Learning Objectives	• Demonstrate secure data storage techniques using proper file/folder permissions, encryption-at-rest, and access controls. • Apply secure configuration checks to identify and correct common storage misconfigurations (e.g., public buckets, weak database access rules). • Document a simple, secure storage plan for a small back-end app/project, including key management and backup considerations.

Session 1

Start of the Lesson (10 minutes)

Quick Retrieval Check: "Secure Storage" in 60 seconds

Individually, answer on paper:

- Define encryption-at-rest in one sentence.
- Give one example of a storage misconfiguration that can cause a data breach.
- List two controls that help protect stored data (e.g., ACLs, least privilege, KMS, backups).
- Why is separating encryption keys from data important?

Pair-share (2 minutes): compare answers and agree on the best response for each item.

Main Task (70 minutes)

Share and Explain (15 minutes)

(Project Check-in)

In pairs/groups, choose ONE data item used by your project/app (e.g., user profile, credentials, transaction logs). Then share:

- Where is it currently stored (local file, database table, cloud storage, etc.)?
- Who should be allowed to access it (roles/permissions)?

Output: 1-page "Data Storage Snapshot" (photo of notes or a short document) to be submitted to GCR.



Let's Learn (20 minutes)

(Lesson and Calibration Guide)

Mini-lesson outline (teacher-facilitated):

- Storage security goals: confidentiality, integrity, availability (CIA) for data at rest.
- Local storage controls: file ownership, permissions (read/write/execute), least privilege, secure defaults.
- Database storage controls: encrypted disks, column/field encryption, strong authentication, network restrictions, protected backups.
- Cloud storage/database: shared responsibility, IAM roles/policies, security groups/firewalls, managed encryption (KMS).
- Common misconfigurations: public buckets, overly broad IAM policies, open database ports, weak admin accounts, unencrypted snapshots/backups.

Let's Practice (35 minutes)

(For practicing the lesson and receiving calibration, feedback, and coaching)

Hands-on Activity: Storage Security Audit

(Local/Offline):

- Create a folder named SECURE_STORAGE_LAB.
- Inside, create three files: public.txt, internal.txt, sensitive.txt.
- Assign access rules (written plan): public.txt = readable by everyone; internal.txt = group-only; sensitive.txt = owner-only + "encrypted" (password-protected archive or documented method).
- Write a 2-3 sentence justification for each rule based on CIA.

Storage Security Checklist (tick Yes/No):

- ☐ Yes ☐ No Data location is identified (file/db/cloud).
- ☐ Yes ☐ No Access roles are defined (admin/user/service).
- ☐ Yes ☐ No Least privilege is applied (no '*' or 'everyone' for sensitive data).
- ☐ Yes ☐ No Encryption-at-rest is enabled for sensitive data and backups/snapshots.
- ☐ Yes ☐ No Secrets/keys are not stored in code or public repos.
- ☐ Yes ☐ No Monitoring/logging is enabled or planned.

Coaching prompt: Present one risk you found, and one fix you proposed. Revise after feedback

Session 2

Start of the Lesson (10 minutes)

Individually, answer on paper:

- Define encryption-at-rest in one sentence.
- Give one example of a storage misconfiguration that can cause a data breach.
- List two controls that help protect stored data (e.g., ACLs, least privilege, KMS, backups).

Pair-share (2 minutes): compare answers and agree on the best response for each item.



Main Task (40 minutes)

Share and Explain (10 minutes)

(Project Check-in)

In pairs/groups, choose ONE data item used by your project/app (e.g., user profile, credentials, transaction logs). Then share:

- What could happen if it becomes public or is modified?

Output: 1-page "Data Storage Snapshot" (photo of notes or a short document) to be submitted to GCR.

Let's Learn (20 minutes)

(Lesson and Calibration Guide)

Mini-lesson outline (teacher-facilitated):

- Calibration checklist (use to self-check work later):
- Access is based on roles (least privilege), not convenience.
- Sensitive fields are encrypted and keys are protected.
- Backups exist and are protected (encrypted + restricted).
- Public access is disabled unless explicitly required and justified.
- Audit/logging is enabled for storage access and changes (where available).

Let's Practice (20 minutes)

(For practicing the lesson and receiving calibration, feedback, and coaching)

Hands-on Activity: Storage Security Audit

Cloud/Online Simulation:

- Given a sample cloud storage/database scenario (teacher provides screenshot/policy snippet), identify at least 3 risks (e.g., public access, wildcard permissions, no encryption).
- Propose corrected settings/policy statements (high level) that enforce least privilege and encryption-at-rest.
- Explain how you would manage keys (who can use the KMS key; how to avoid hardcoding secrets).
- Submission: (1) screenshot/photo of output + (2) completed Storage Security Checklist.

Storage Security Checklist (tick Yes/No):

- ☐ Yes ☐ No Data location is identified (file/db/cloud).
- ☐ Yes ☐ No Access roles are defined (admin/user/service).
- ☐ Yes ☐ No Least privilege is applied (no '*' or 'everyone' for sensitive data).
- ☐ Yes ☐ No Encryption-at-rest is enabled for sensitive data and backups/snapshots.
- ☐ Yes ☐ No Secrets/keys are not stored in code or public repos.
- ☐ Yes ☐ No Monitoring/logging is enabled or planned.

Coaching prompt: Present one risk you found, and one fix you proposed. Revise after feedback

References	Student Feedback Form
• Information Assurance and Security 1 course outline: Module 9 - Secure Data Storage (file permissions, encrypted databases, cloud database security, storage misconfiguration risks). • Instructor/NetAcad readings and demos on OS file permissions and database security. • OWASP guidance/handouts on secure storage configuration and secrets management.	





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Think Back and Think About (10 minutes)

Answer briefly (3-5 sentences each):

- What was the most common storage risk you saw today (or in the scenario)? Why is it risky?

- Which control (permissions, encryption, IAM, backups, logging) is most important for your project? Explain.

- If you were a user, what would you expect to do to protect your stored data?

Wrap Up (20 minutes)

Please use the space below (or the space at the back) to write your answers. Questions will be provided by the teacher.

